DATA ITEM DESCRIPTION

Title: ACOUSTIC AND SONIC FATIGUE ANALYSIS REPORT

Number: DI-ENVR-82358
Approval Date: 20210917
AMSC Number: F10267
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFS)
Project Number: ENVR-2021-002
Applicable Forms: N/A

Use/Relationship: The Acoustic and Sonic Fatigue Analysis Report describes the approach for establishing the acoustic environments of structures and equipment and predicting the acoustic strength and fatigue life of airframe structures subjected to these environmental loads. This analysis report supports the design of the air vehicle.

a. This Data Item Description (DID) is applicable to any contract for which an air vehicle is under development, a modification is made, or an Environmental Criteria Document (ECD) is required.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in section 5 of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)*. (Copies of this document are available online at https://quicksearch.dla.mil.)

c. This DID supersedes DI-ENVR-82121, *Vibration and Acoustic Analyses Reports – Air Vehicle*. (Copies of this document are available online at https://quicksearch.dla.mil.)

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Acoustic and Sonic Fatigue Analysis Report shall be in the contractor's format.

3. Content. The Acoustic and Sonic Fatigue Analysis Report shall describe the contractor's analysis associated with the design for acoustic loading environments. The report shall include the following:

a. Acoustic Loads and Environmental Analysis, to include:

(1) Approach used to establish the acoustic loading and environmental exposure of the structure and equipment for ground and flight operations.

(2) Description of all sources of acoustic loads, including: operation and efflux of propellers, rotors, engines, and thrust reversers; boundary layers; wakes; oscillating shocks and other aerodynamic disturbances; Mach effects; ground operations; cavity and inlet noise; missile and rocket launches; and gunfire.

(3) Substantiation of any design provisions employed to control acoustic load levels.

(4) Predicted levels for harmonic, transient, and random acoustic load sources or combinations of sources with graphs, calculations, data tables, etc., in sufficient detail to derive the exposure of the system by physical location and with sufficient frequency resolution to evaluate local design features.

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(5) Description of the acoustic loading with graphs and tables that define the loading as a function of frequency and exposure time.

(6) Relevant service environment details that contribute to the acoustic loading and resulting service life, including: temperature, humidity levels, and cabin pressurization.

(7) Description of analysis tools and analytical models.

(8) Usage data in terms of mission types, mission mix, storage, and transportation used to develop the repeated loads and environment spectra for use in the service life analysis.

(9) Descriptions of the theories, methods, and assumptions used to modify acoustic load levels to support the accelerated testing procedures described in the ECD, if applicable.

b. Sonic Fatigue Service Life Analysis, to include:

(1) Sonic fatigue analysis methodology and validation results used to design the air vehicle structure and equipment subjected to acoustic loading.

(2) Description of analysis tools and analytical models.

(3) Fatigue life predictions of all acoustic sensitive structures and all supporting calculations and data, such as loads-to-stress transfer functions and material properties.

(4) Usage data in terms of mission types, mission mix, storage, and transportation used to develop the repeated loads and environment spectra for use in the service life analysis.

c. Analysis Correlation, to include:

(1) Comparison of pre-test analysis predictions to the acoustic, environmental, and sonic fatigue test data.

(2) Analysis results and correlation of the analytical tools used in the acoustic, environmental, and sonic fatigue analysis.

(3) Comparison of the acoustic, environmental, and sonic fatigue test results with previous design levels and qualification test requirements.

d. Distribution Statement (i.e., A, B, C, D, E, etc.) in accordance with Department of Defense Instruction (DoDI) 5230.24, *Distribution Statements on Technical Documents*, Enclosure 4. (Copies of this document are available online at www.esd.whs.mil/DD.)

e. The applicable classification marking, as necessary, in accordance with Department of Defense Manual (DoDM) 5200.01, Volume 2, *DoD Information Security Program: Marking of Information*. (Copies of this document are available online at www.esd.whs.mil/DD.)

End of DI-ENVR-82358.

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